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Co-electroreduction of CO and glyoxal promotes C_3 products

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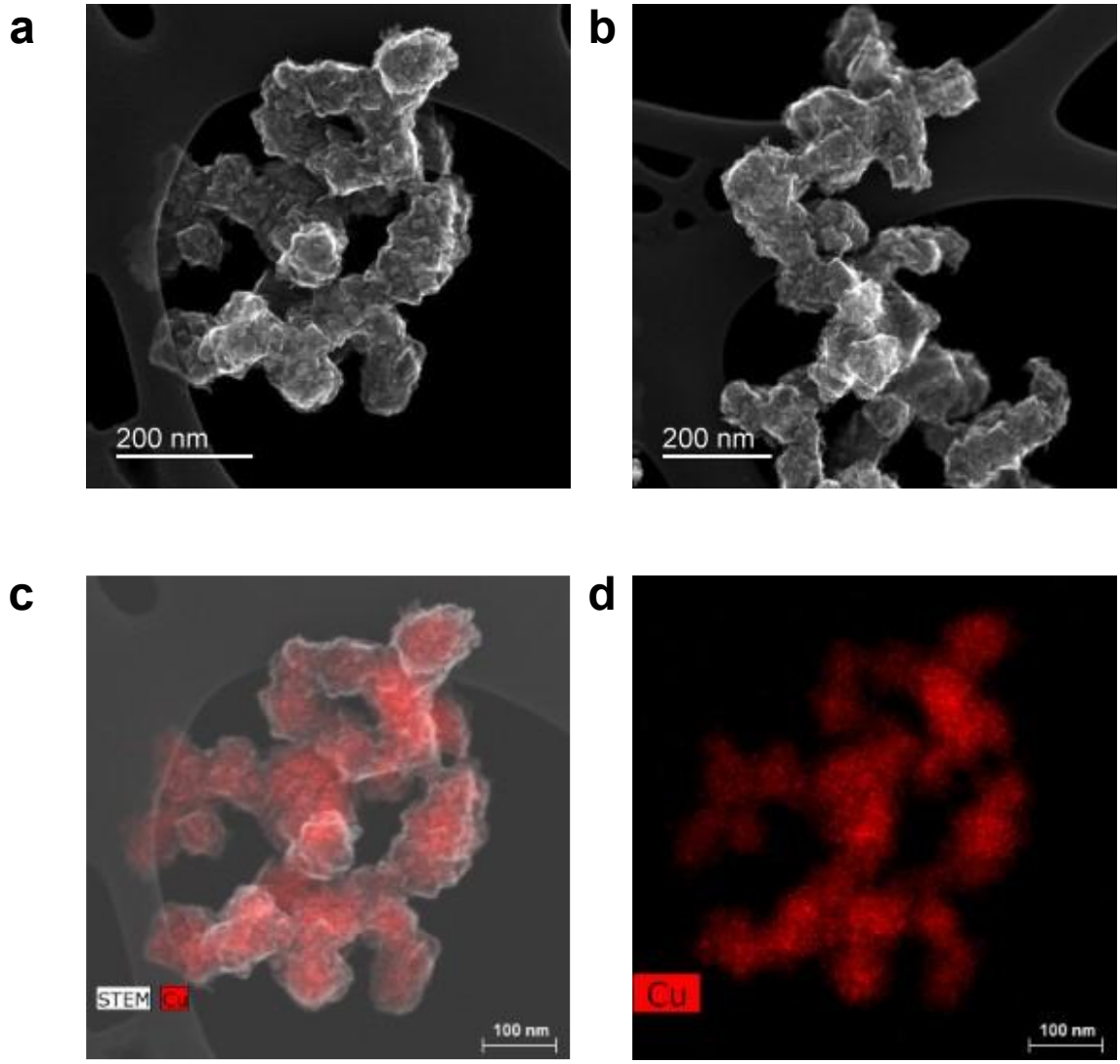
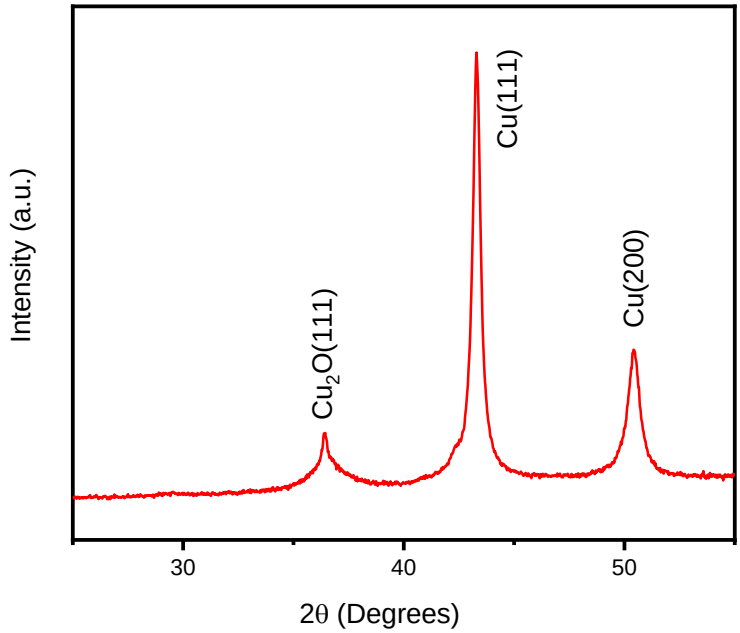
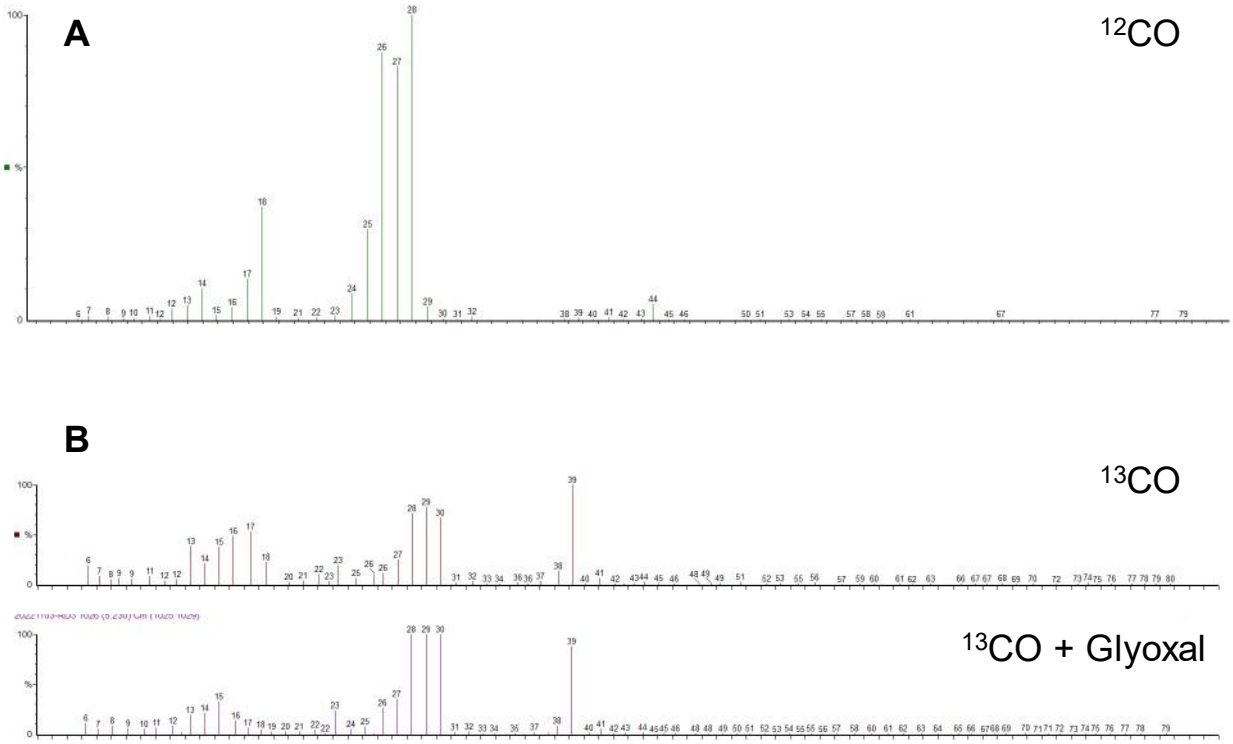
Supplementary Table 1. C_{3+} performance summary **32**



Supplementary Fig. 1. a, b Transmission electron microscopy (TEM) of copper nanoparticles. **c, d** Energy dispersive X-ray spectroscopy (EDS) of copper nanoparticles.



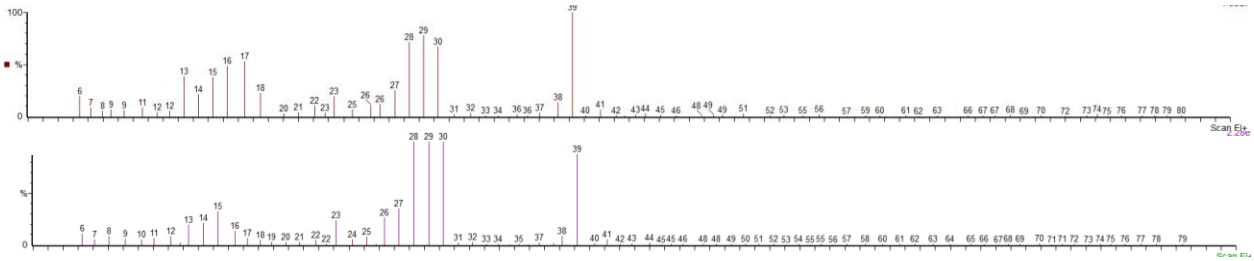
Supplementary Fig. 2. X-ray diffraction (XRD) of the copper nanoparticles.



Supplementary Fig. 3. The mass fragmentation patterns of ethylene produced from (**A**) un-labelled CO and (**B**) labelled ^{13}CO with and without ^{12}C-glyoxal. The peaks at 28, 27, 26 and 25 m/z correspond to $[C_2H_4]^+$, $[C_2H_3]^+$, $[C_2H_2]^+$ and $[C_2H]^+$ fragments of ethylene respectively.

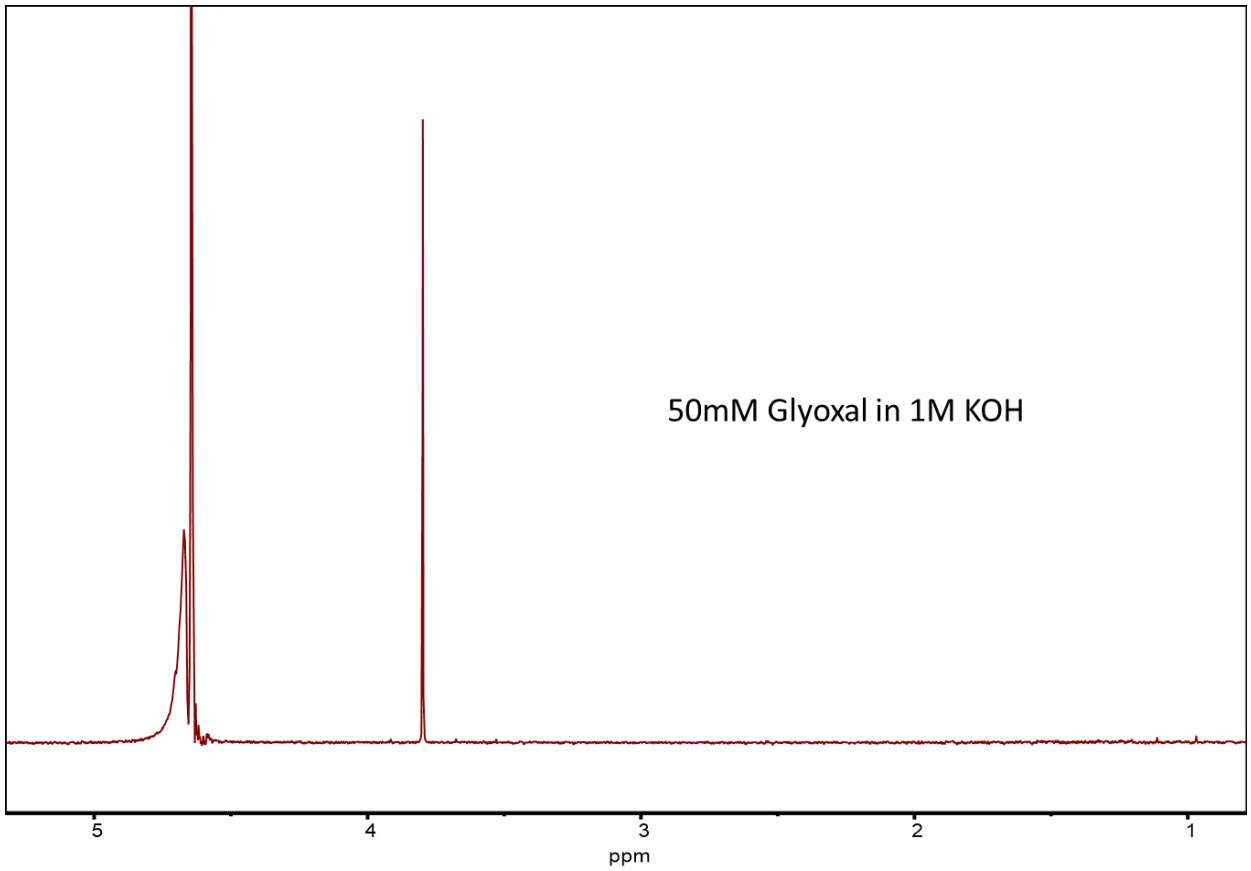
Note: The m/z=18 peak corresponds to residual water.

The fragment corresponding to m/z=39 could not be assigned to any major product found in the reaction. Further experiments showed its presence in subsequent samples using a different labelled CO cylinder, indicating it may be contamination from the cylinder or artifact.

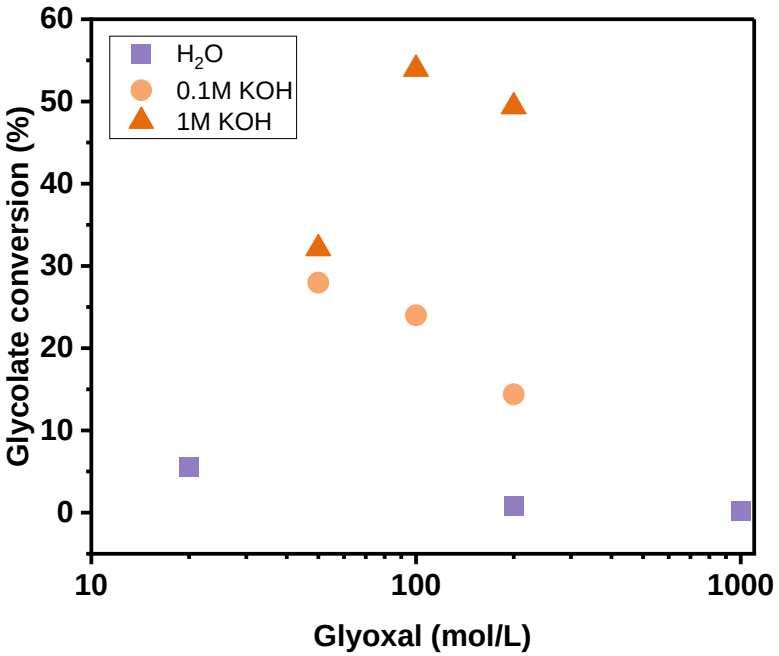


Supplementary Fig. 4. Two samples taken from different tests and different ^{13}C-CO cylinders (Sigma Aldrich) during CORR.

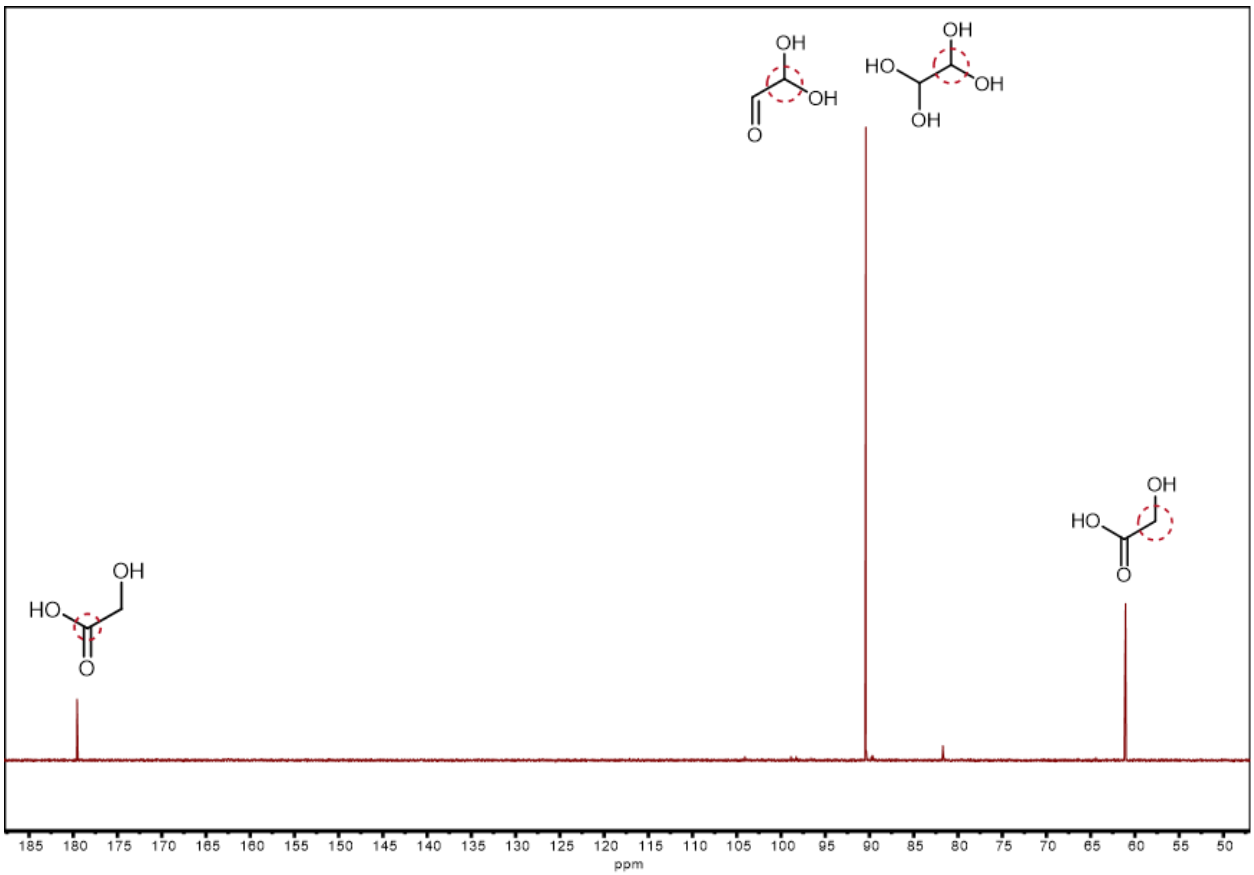
Higher intensities of this peak (m/z=39) in Figure S1B are attributed to the lack of corresponding contamination in non-labelled CO (Figure S1A). The water contamination artifact (m/z=18) signal varies with sample and humidity present at different lab conditions (Figure S1).



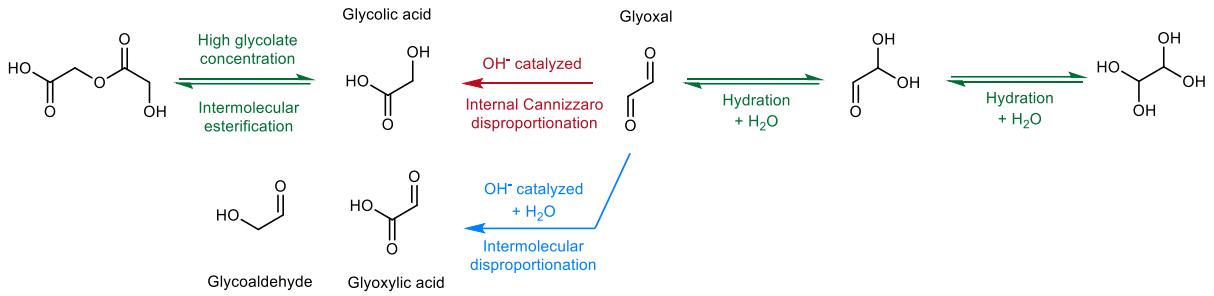
Supplementary Fig. 5. NMR spectrum of liquid products obatined on Cu-Gly electrode under CORR without application of current. Presence of the glycolate peak at 3.8 ppm, and overlap of glyoxal peak (4.8 ppm) with remaining water (4.5 – 4.9 ppm).



Supplementary Fig. 6. Fraction of glyoxal converted to glycolate (glycolate conversion) in different electrolytes using various initial glyoxal concentrations. Samples were analyzed after at least 1 hour to ensure equilibrium had been reached.



Supplementary Fig. 7. ^{13}C-NMR of glyoxal in alkaline electrolyte (KOH solution). The solution was left to equilibrate for 2-4 hours, a similar timescale in which all electrolysis experiments have been performed in.



Supplementary Fig. 8. Transformations of glyoxal in an alkaline electrolyte. Red and blue indicate irreversible reactions, while green indicates equilibrated reactions.

Supplementary Note 1. Computational framework.

We employed the Vienna Ab initio Simulation Package (VASP) for all geometry optimizations and electronic structure calculations.[1-2] The interaction between valence electrons and ionic cores was described using the projector augmented-wave (PAW) method,[2] ensuring a high level of accuracy by leveraging a frozen-core all-electron framework. To capture van der Waals interactions critical to describing surface-adsorbate interactions, we applied Grimme's dispersion correction to the Perdew-Burke-Ernzerhof (PBE) functional. [3] Atomistic relaxations were conducted using the conjugate gradient method, with a stringent energy convergence threshold of 0.0001 eV $Å^{-1}$ and a plane-wave energy cutoff set to 520 eV. Structural optimizations were performed using the PBE-D3 exchange-correlation functional, accounting for both lattice parameters and electronic properties. [4-5] Copper (Cu) in the Fm3m space group was modeled using the Atomic Simulation Environment (ASE) framework.[6] Full relaxation of ionic positions, cell shape, and volume was carried out to identify the ground-state geometry. For the bulk Cu calculations, we employed a Γ-centered 12×12×12 k-point mesh to ensure accurate sampling of the Brillouin zone. From the optimized bulk structures, Cu (100) surfaces were generated by cleaving along the (100) plane, producing a slab model with four atomic layers. A supercell consisting of 144 atoms (6×6×1 slab configuration) was constructed, with in-plane lattice constants a, b = 15.34 Å, and a vacuum spacing of 30 Å introduced along the z-direction to mitigate interactions between periodic images. K-point sampling for the Cu(100) surface was reduced to a Γ-centered 2×2×1 mesh during surface optimization, which was sufficient to capture surface electronic effects accurately.

Surface Gibbs Free Energy Calculations for Glycolate Adsorption and CO Reduction on Cu(100):

To investigate the adsorption of glycolate on the Cu(100), we systematically evaluated several possible adsorption sites, including top, bridge, and hollow configurations, identifying the most energetically favorable binding geometry **(Supplementary Fig. 9)**. The glycolate molecule exhibits a strong binding affinity to the Cu (100) surface, with a binding energy of -3.43 eV. It preferentially binds at the bridge side of the surface, coordinating through its oxygen atoms. In addition, we also consider the

adsorption of CO (gas) molecule the pristine and pre-adsorbed glycolate on the Cu (100). Considering water as an implicit solvent, our calculations show that the CO adsorption energy on the pristine Cu (100) surface is -0.54 eV, which is consistent with previous reports.[7] Notably, CO adsorption on the glycolate pre-adsorbed Cu (100) surface is significantly stronger, with an energy of approximately -1.73 eV. Solvation effects were incorporated using the implicit Poisson-Boltzmann solvation model implemented in **vaspsol++**, with a dielectric constant of water set to 80, mimicking an aqueous environment[8, 9]. The free energy changes (ΔG) for CORR intermediates were calculated following the methodology established by Nørskov and co-workers[10,11]. This approach considers contributions from the adsorption energy, zero-point energy corrections, and entropy effects, providing a robust framework for evaluating reaction energetics at the atomic scale.

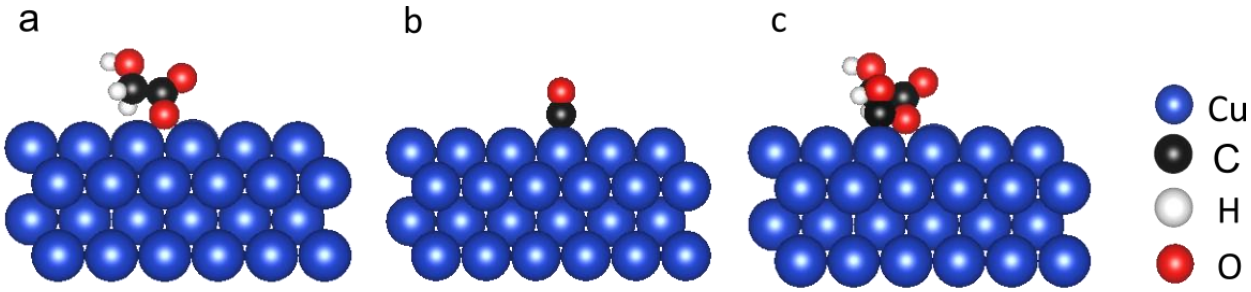
$$\Delta G = E_{\mathrm{ads}} + \Delta E_{\mathrm{ZPE}} - T\Delta S + \Delta G_{\mathrm{pH}} + \Delta G_{\mathrm{U}} \quad \text{(S1)}$$

The adsorption energy E_{ads} was determined from density functional theory (DFT) calculations, with corrections applied to account for vibrational and entropic contributions. The zero-point energy difference ΔE_{ZPE} and the entropy (ΔS) were included in the free energy calculations. The system temperature was fixed at 298.15 K. Zero-point energy corrections were computed by summing contributions from all vibrational modes ($E_{\mathrm{ZPE}} = \frac{1}{2}\sum h\nu$), where ν represents the vibrational frequencies. Entropic contributions for gas-phase species such as CO, CO_2, H_2, and H_2O were extracted from the NIST database[12]. The influence of pH on the reaction energetics was included using the term $\Delta G_{\mathrm{pH}} = 0.059 \times \mathrm{pH}$, which reflects the change in free energy associated with variations in the proton concentration. Additionally, the effect of an applied electrode potential was incorporated for all states involving single-electron transfer steps, with the potential-dependent energy shift calculated as $\Delta G_{\mathrm{U}} = -eU$, where U is the electrode potential relative to the standard hydrogen electrode (SHE). These corrections provide a comprehensive thermodynamic framework for evaluating the energetics of the CORR under electrochemical conditions. For the glycolate pre-adsorbed system, all Gibbs free energy calculations were performed in the charged state, incorporating water as an

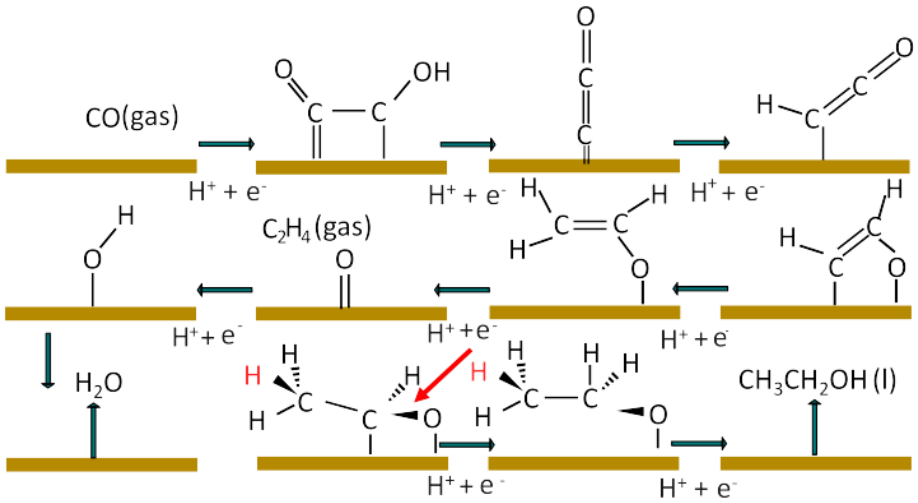
implicit solvent to accurately capture solvation effects. The mechanistic framework used is show in **Supplementary Fig. 10**, along with the resulting energetic landscape when Cu(100) and Cu(100)-glyoclate are used as the basis. **(Supplementary Fig. 11**)

Effect of Glycolate Coverage on CO Adsorption and C_{2+} Product Selectivity on Cu(100):

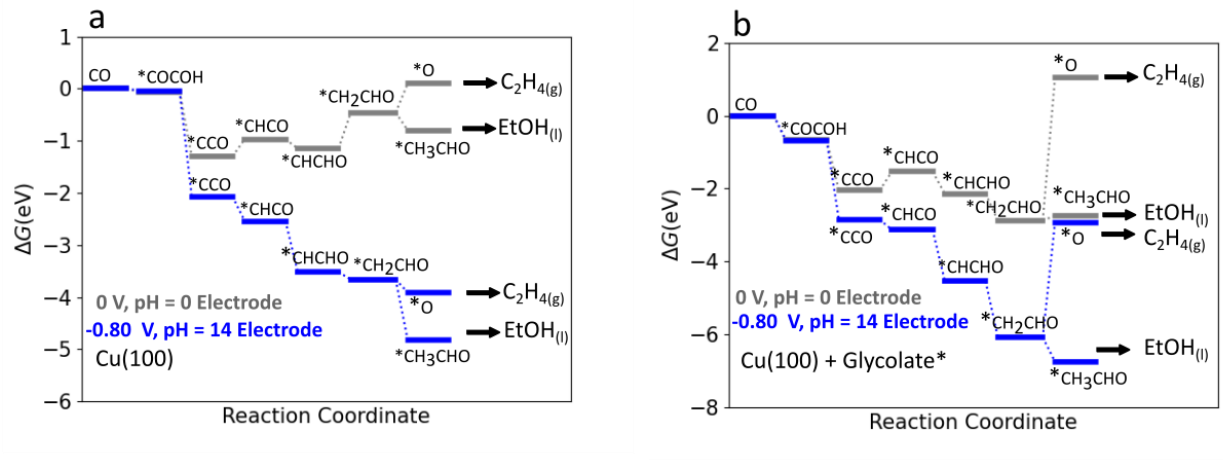
To investigate the effect of glycolate coverage on *CO adsorption and the selectivity of C_{2+} product formation, we first consider how the adsorption configurations of 2, 4, and 6 glycolate molecules would behave on a clean Cu (100) surface **(Supplementary Fig. 12)**. We used the ACAT Python-based package to generate all possible configurations for glycolate molecule coverage[13]. Specifically, we explored adsorption at various sites, including top, bridge, and four-fold hollow sites, as well as combinations of these sites (top and bridge, top and four-fold hollow, and bridge and four-fold hollow) to identify the most stable configuration for each coverage level. Building on the most stable glycolate adsorption configurations for each coverage, we then examined *CO adsorption on the glycolate-pre-adsorbed Cu(100) surface. Our calculations reveal that *CO adsorption is mildly stabilized **(Supplementary Fig. 13)**. To investigate how glycolate coverage influences the selectivity for C_{2+} product formation, we focused on two key intermediates: *CCH and *HOCHCH, whose protonation leads to the formation of ethylene (C_2H_4) and ethanol, respectively. Our calculations show that the adsorption of both *CCH and *HOCHCH is mildly stabilized, with HOCHCH* having a lower formation gibbs free energy under all glycolate coverages **(Supplementary Fig. 14)**. This suggests that glycolate is unlikely to considerably shift the equilibrium of the selectivity determining step. The most stable configurations for the adsorption of both intermediates at each coverage are depicted in **Supplementary Fig. 15**. Both CO* adsorption studies and the selectivity determining step comparison suggest that glycolate has a minimal effect on the C_{2+} product distribution, corroborating what we observe experimentally.



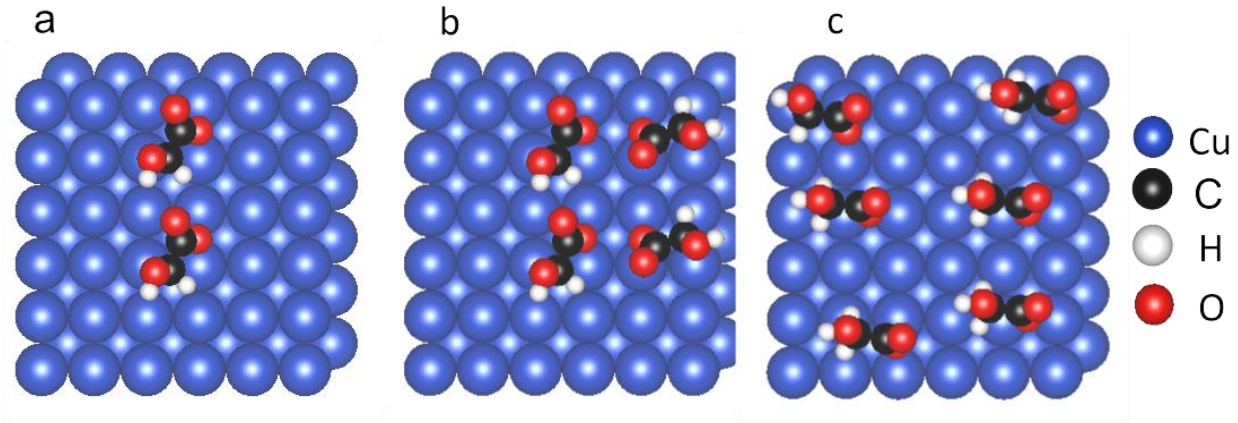
Supplementary Fig. 9. Chemical structures illustrating (a) glycolate adsorbed on the Cu (100) surface, (b) CO adsorbed on the pristine Cu surface, and (c) CO adsorbed on the glycolate pre-adsorbed Cu surface. The light blue, black, red, and white spheres correspond to Cu, C, O, and H atoms, respectively.



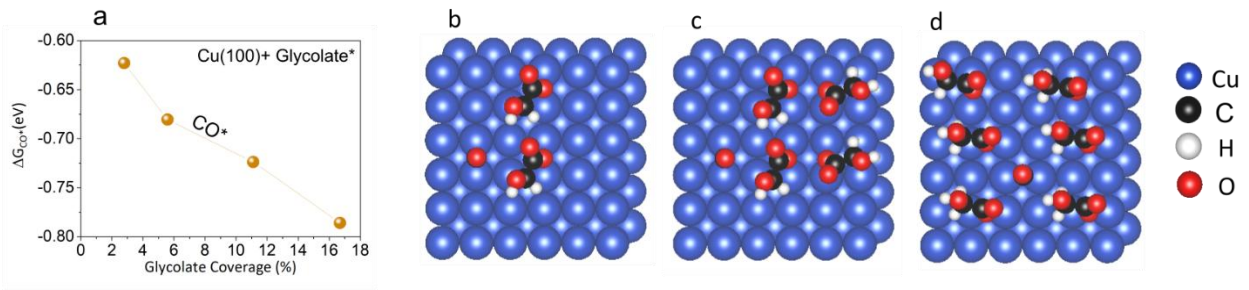
Supplementary Fig. 10. Complete reaction pathway illustrating the free energy profiles for CO(g) reduction to ethanol (EtOH) and ethylene (C_2H_4) on a clean and glycolate-pre-adsorbed Cu(100) surface at pH = 14. Calculations were performed under two electrode potentials: 0 V and -0.80 V vs. SHE.



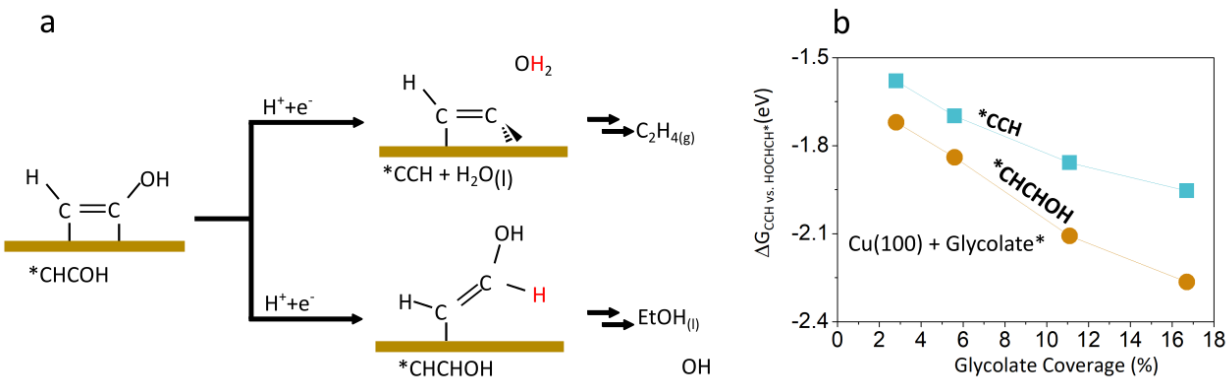
Supplementary Fig. 11. Gibbs free energy profiles for the CO(g) reduction reaction to ethanol (EtOH) and ethylene (C_2H_4) on the Cu (100) surface at pH 14, computed under two electrode potentials: 0 V and -0.80 V vs. SHE. (a) Profiles for the pristine Cu (100) surface, showing the reaction energetics for each step. (b) Profiles for the Cu (100) surface pre-adsorbed with glycolate, highlighting the influence of glycolate adsorption on the reaction pathway. The label "EtOH" indicates the formation of liquid ethanol as a product.



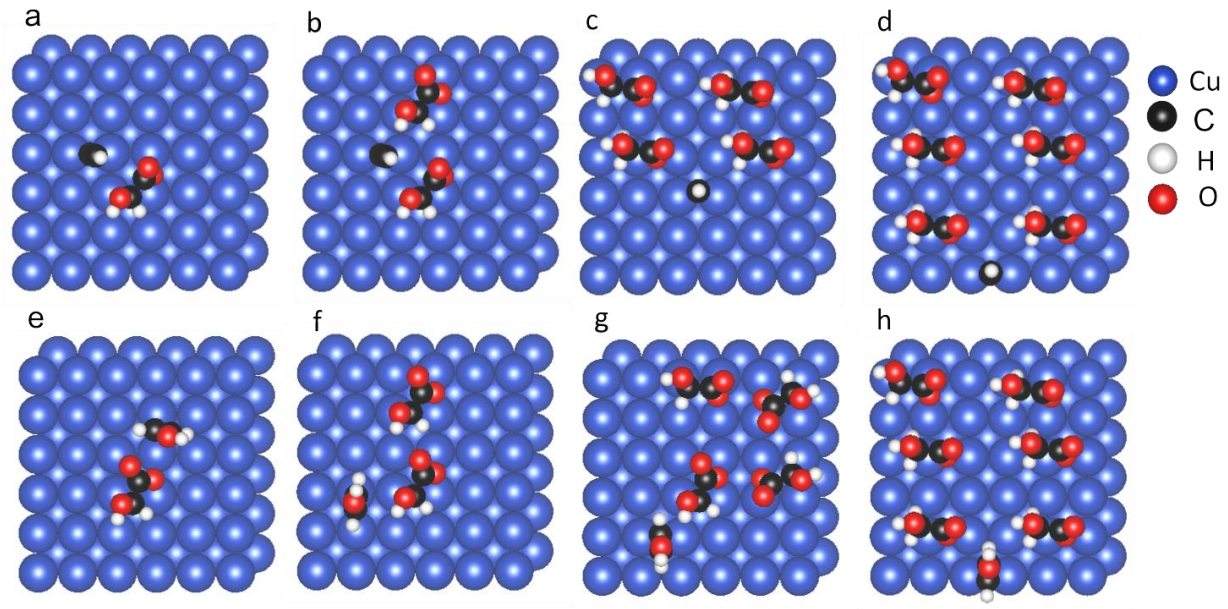
Supplementary Fig. 12. Optimized chemical structures of the most stable glycolate adsorption configurations on the Cu(100) surface at varying coverages: (a) 2-glycolate, (b) 4-glycolate, and (c) 6-glycolate. Light blue, black, red, and white spheres represent Cu, C, O, and H atoms, respectively.



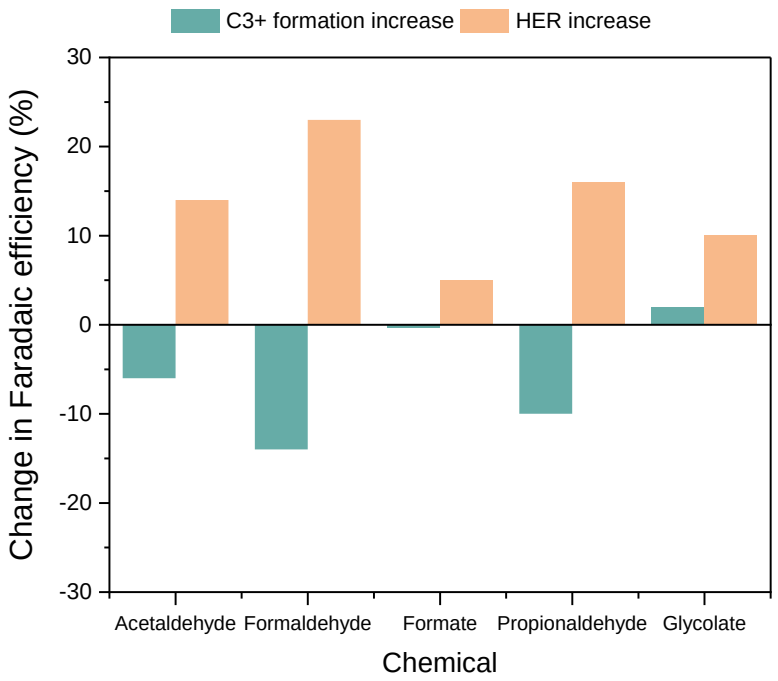
Supplementary Fig. 13. (a) Adsorption energy of CO* on the glycolate-pre-adsorbed Cu(100) surface as a function of glycolate coverage. (b-d) Optimized stable configurations of CO* at varying glycolate coverages on Cu(111). Light blue, black, red, and white spheres represent Cu, C, O, and H atoms, respectively.



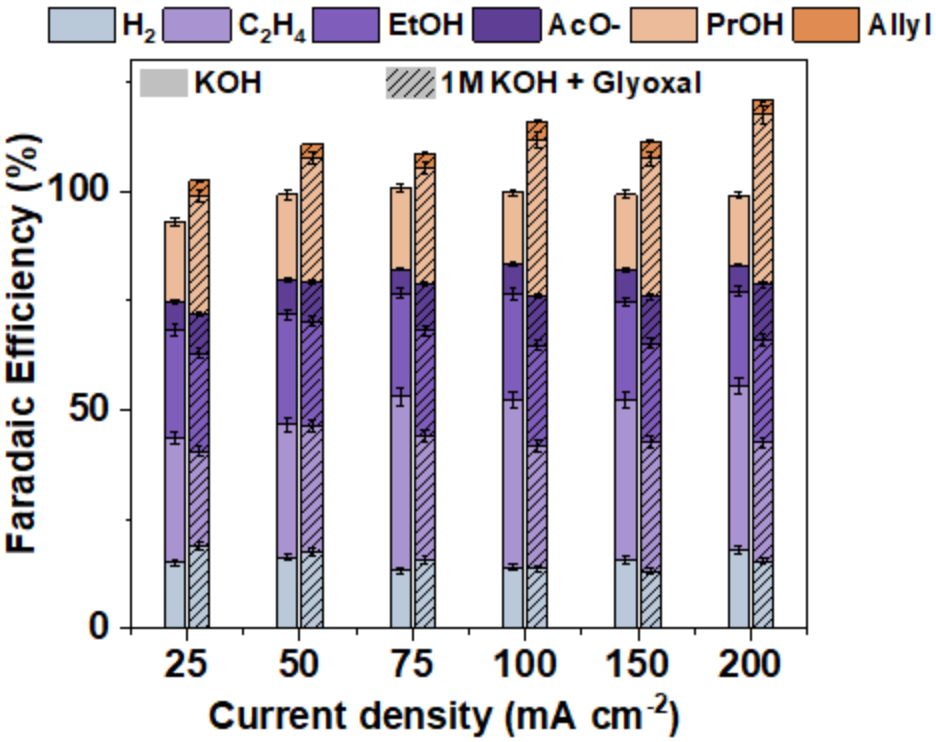
Supplementary Fig. 14. (a) The selectivity-determining step in CO reduction is proposed to be the dehydrogenation of HCCOH (favoring ethylene formation) or its protonation (favoring ethanol formation). The Gibbs free energy barriers of these two pathways serve as a key descriptor for ethylene/ethanol selectivity. (b) Gibbs free energy of CCH* and HOCHCH* as a function of glycolate coverage on the Cu(100) surface.



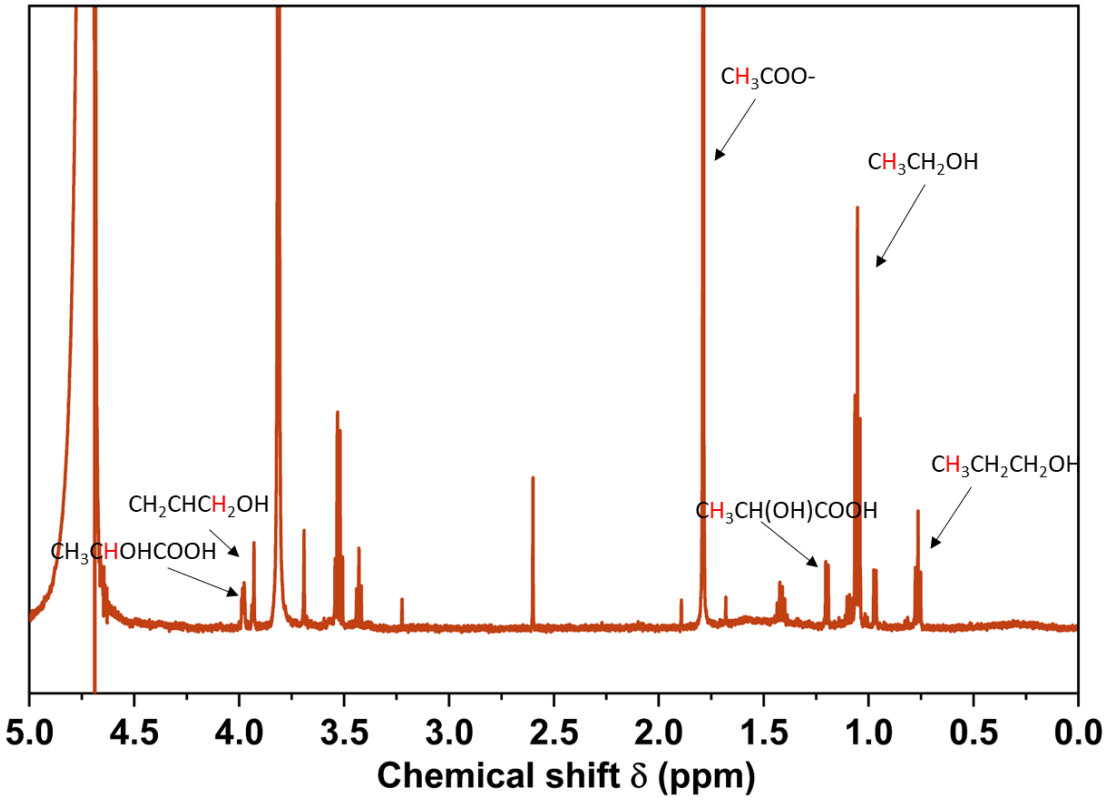
Supplementary Fig. 15. The most stable configurations of (a-d) CCH* and (e-f) HOCHCH* intermediate at varying coverages of pre-adsorbed glycolate on the Cu(100) surface. The light blue, black, red, and white spheres represent Cu, C, O, and H atoms, respectively.



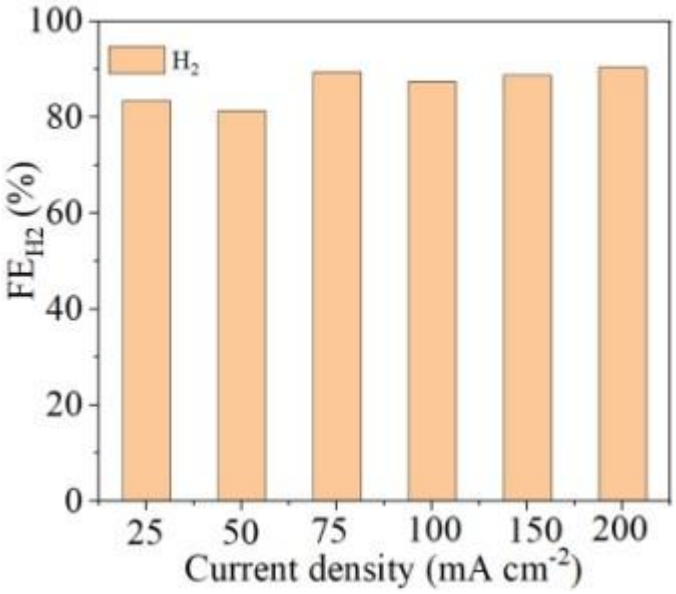
Supplementary Fig. 16. Change in C_3 formation FE and HER FE upon addition of different C_1-C_3 organics into the catholyte (1M KOH + 100 mM organic compound).



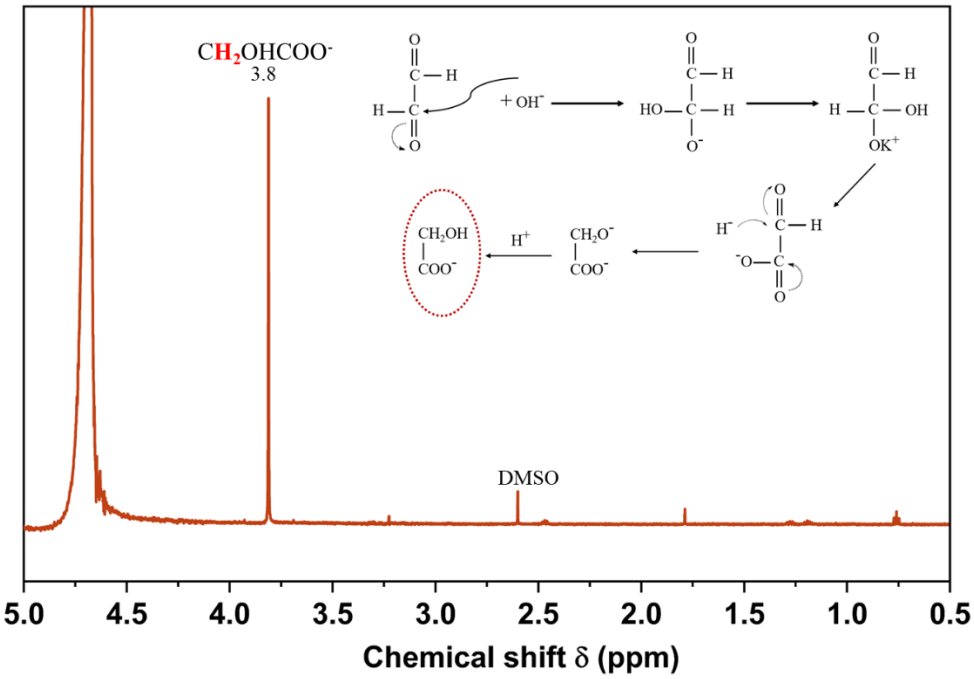
Supplementary Fig. 17. Product distribution under different current densities for Cu-Gly and Cu samples. Values are means and error bars indicate standard deviation of three independent samples.



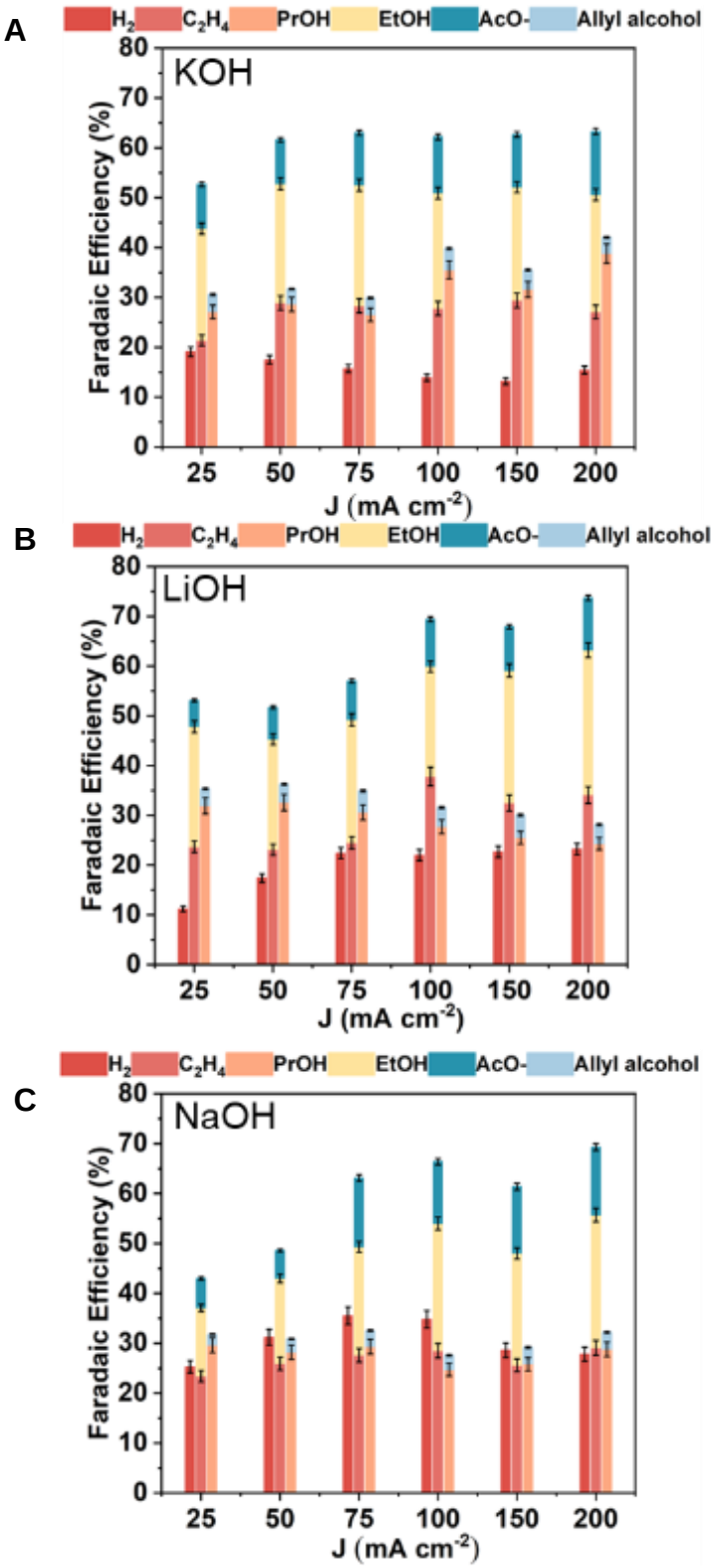
Supplementary Fig. 18. NMR spectrum of liquid products obtained on Cu-Gly under CORR conditions with application of current, with notable presence of allyl alcohol and lactic acid as new C_3 products.



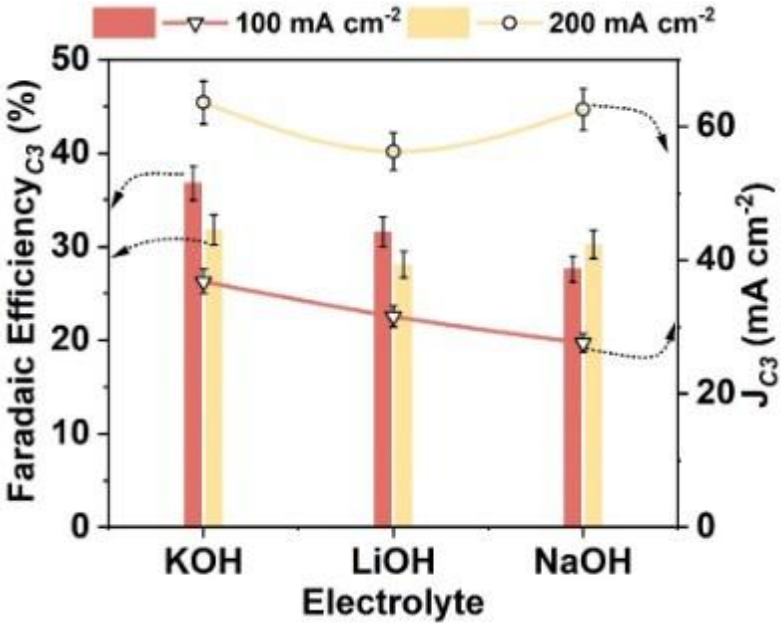
Supplementary Fig. 19. FE of products as obtained on Cu-Gly electrode under N_2 atmosphere. Only H_2 was detected as the gaseous product.



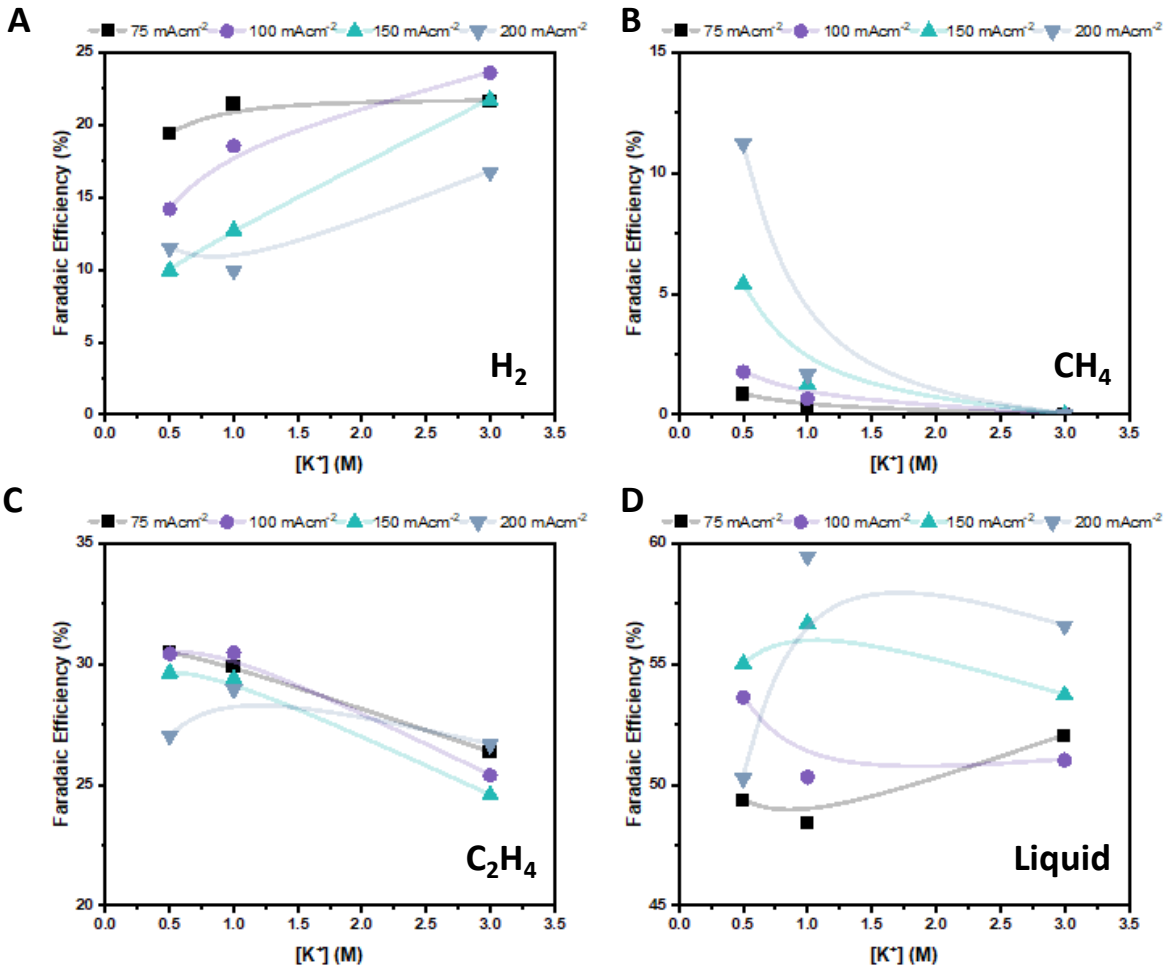
Supplementary Fig. 20. NMR spectrum of liquid products (catholyte) obtained on Cu-Gly electrode under N_2 atmosphere.



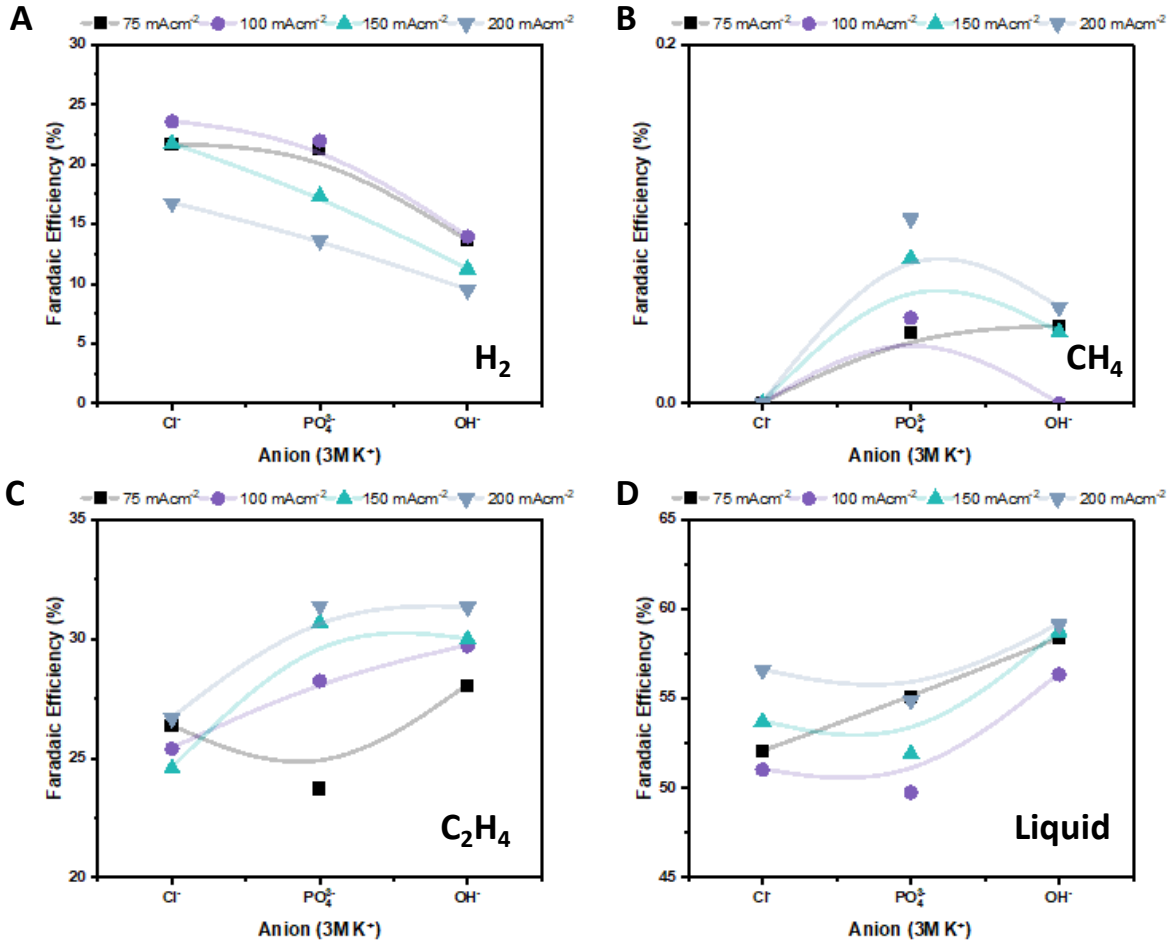
Supplementary Fig. 21. CORR product distribution of Cu-Gly electrodes in (**A**) 1 M KOH solution, (**B**) 1M LiOH solution and (**C**) 1 M NaOH solution. Values are means and error bars indicate standard deviation of three independent samples.



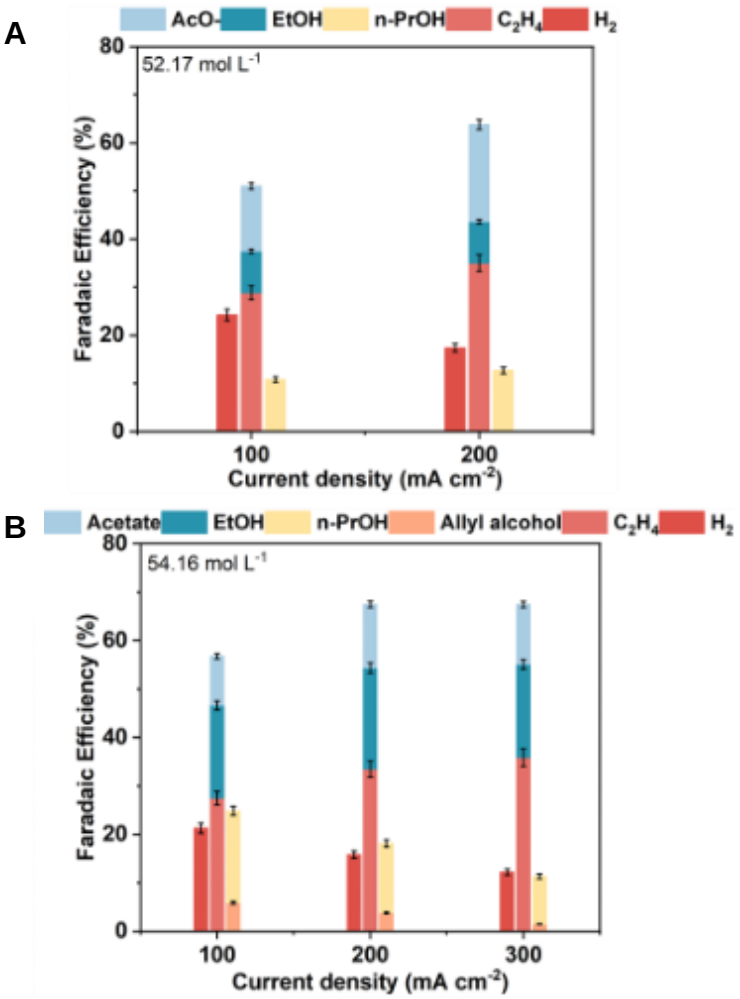
Supplementary Fig. 22. C_3 FEs and partial C_3 current densities of Cu-Gly (50mM) using various electrolytes (1M) at current densities of 100 and 200 mA cm^{-2}. Values are means and error bars indicate standard deviation of three independent samples.



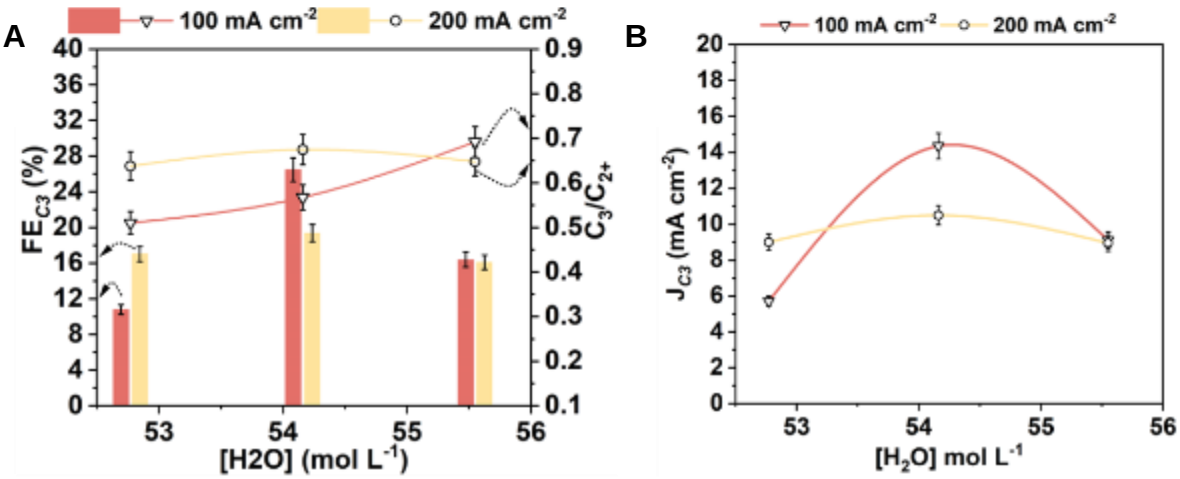
Supplementary Fig. 23. (**A**) Hydrogen, (**B**) methane, (**C**) ethylene and (**D**) liquid product FEs of Cu-Gly (50mM) using KCl catholyte at various concentrations and current densities.



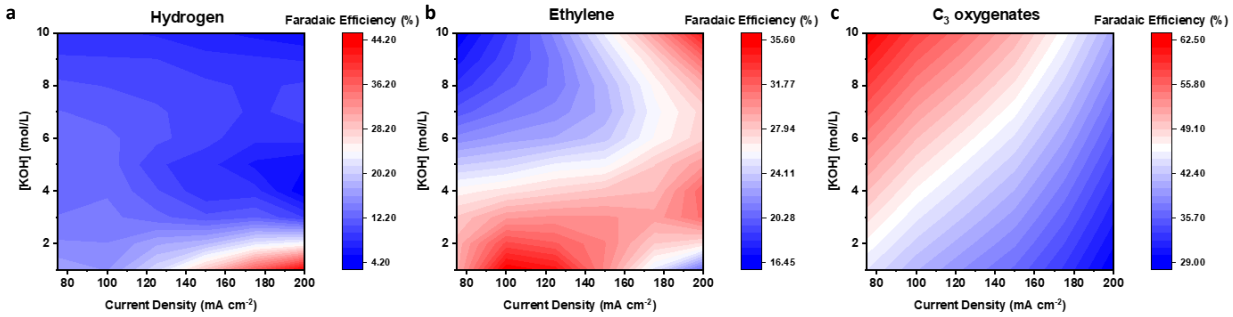
Supplementary Fig. 24. (**A**) Hydrogen, (**B**) methane, (**C**) ethylene and (**D**) liquid product FEs of Cu-Gly (50mM) using different catholyte salts containing 3 M K^+ concentration e.g., 1 M K_3PO_4 and 3 M KCl and KOH.



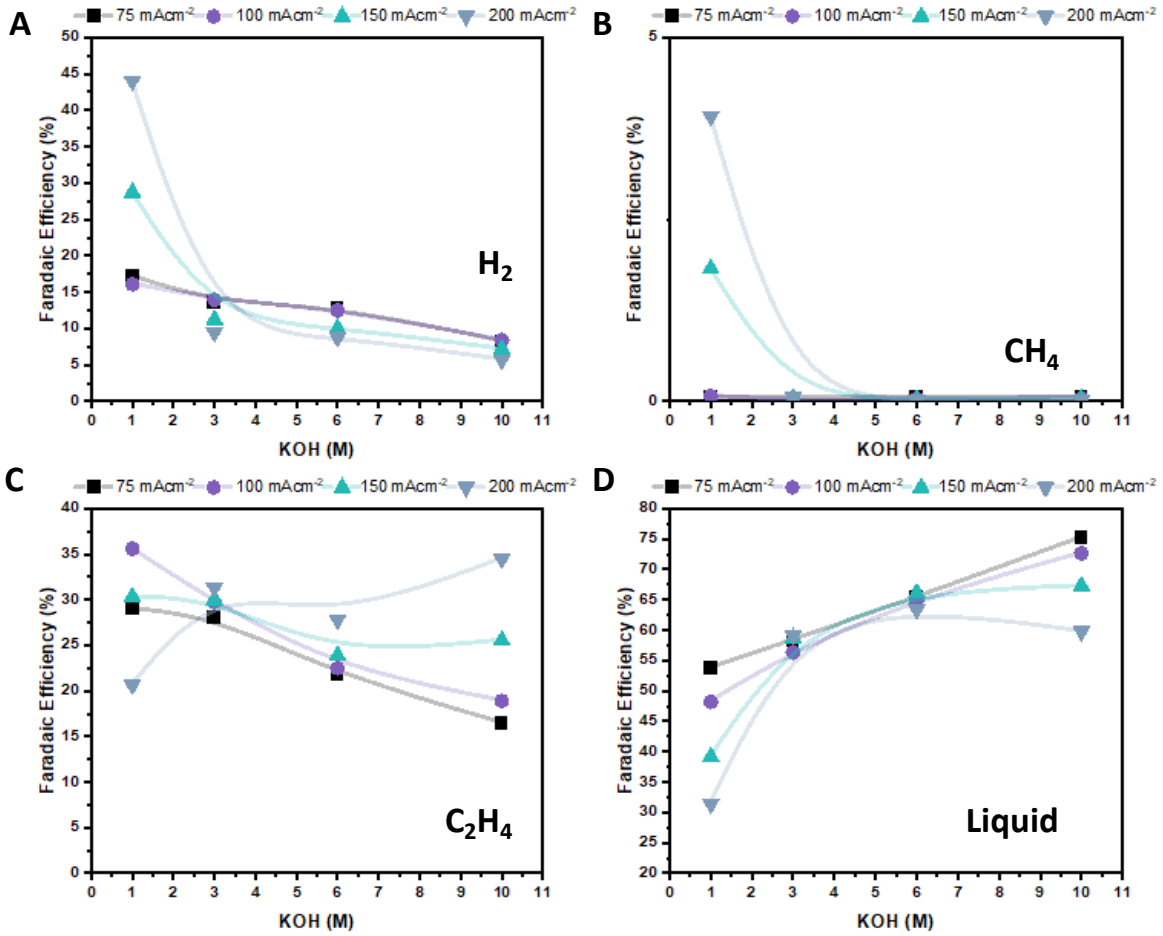
Supplementary Fig. 25. CORR product distribution of CORR on Cu electrodes in 1M KOH + DMSO under varying water concentration. (**A**) 52.17 mol L^{-1} and (**B**) 54.16 mol L^{-1}. Values are means and error bars indicate standard deviation of three independent samples.



Supplementary Fig. 26. C_3 (**A**) FE and (**B**) partial current density of Cu electrodes under CORR conditions in 1M KOH + DMSO. Values are means and error bars indicate standard deviation of three independent samples.



Supplementary Fig. 27. (**g**) Hydrogen (**h**) ethylene (**i**) C_3 oxygenate FEs at varying current densities and KOH concentrations at 1 bar CO partical pressure and electrolytes containing 50mM glyoxal.



Supplementary Fig. 28. (**A**) Hydrogen, (**B**) methane, (**C**) ethylene and (**D**) liquid product FEs in different KOH concentration catholytes containing 50 mM glyoxal.

Supplementary Table 1. Performance comparison toward C_{3+} oxygenates in prior studies vs. the present one:

Catalyst (Cu)	CO_2 or CO	C_3 products	FE_{C3} ($FE_{n\text{-}PrOH}$)	J_{tot} (mA cm^{-2})	Ref
Cu NP	**CO**	**Allyl alcohol, Lactic acid, n-Propanol**	**~63% (~53%)**	**75**	**This work**
Cu/Cu2O@N-doped graphene	CO_2	n-Propanol	8%	20	J. CO_2 Utilization, V. 50, 2021, 101594
Cu NP ensemble	CO	Allyl alcohol, Acetone, n-Propanol	~7%	10	PNAS 114 (40), 10560-10565.
Cu nano-cavity	CO	n-Propanol	21%	40	Nat Catal 1, 946–951 (2018).
Cu NP (<100 nm)	CO	n-Propanol	25%	30	Electrochemica acta., V 307, 2019, 164-175
Cu_2O multi-hollow	CO	n-Propanol	~16%	60	Adv. Eng. Mat, V 12, 36, 2022, 2202054
Catalyst (other)	**CO_2 or CO**	**C_3 products**	**FE_{C3} ($FE_{n\text{-}PrOH}$)**	J_{tot} (mA cm^{-2})	**Ref**
Ag-Cu_3N	CO	n-propanol	45%	150	Angewandte Chemie International Edition, 62(49)
Ag-Cu	CO_2 (supersaturated)	Isopropyl alcohol	55%	100	Nat Cat 6, 319-331 (2023)
Pb-rich Cu	CO	n-propanol	46%	80	Nat Commun 14, 4882 (2023)
CuS	CO_2	n-Propanol	15%	10	Nat Commun 12, 1580 (2021).
Cu-Ag	CO	n-Propanol	32%	5	Nat Commun 10, 5186 (2019).
$Ni^{\delta+}$	CO	C_3-C_6 hydrocarbon	7%	12	Nat Catal 5, 545–554 (2022).

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